

**STRULE AUTOMATION**

Industrial Controls · Bay Area & Northern California

TRAINING PRIMER

Alarms and Interlocks: What They're For, and Why You Never Defeat One

I've spent a lot of years standing at motor control centers and operator screens with maintenance folks, and I've learned that most of the trouble around alarms and interlocks isn't mechanical. It's vocabulary. People use five words for things that are genuinely different, the screen lights up like a Christmas tree at 3 a.m., and eventually somebody reaches for a jumper wire because "that alarm's always going off anyway." This primer is my attempt to hand you the words, clearly, and then walk through the one line you never cross. If you run a materials recovery facility (MRF), an aggregate plant, or an organics/anaerobic digestion (AD) site, this is for you.

Why this matters on the floor

The controls on your plant are trying to do two jobs at once. One job is keeping product moving. The other job is keeping people and equipment from getting hurt when something goes wrong. Those two jobs share the same wires, the same programmable logic controller (PLC — the industrial computer that runs the sequence), and the same screens. When you can't tell which job a given signal is doing, you can't make a good decision about it. You'll ignore something that was begging you to act, or you'll defeat something that was the only thing standing between a conveyor and somebody's hand.

So let's separate the ideas. They feel similar. They are not.

Five words, five different jobs

An alarm tells a human to act

An alarm (an annunciated abnormal condition asking for a response) is a message. That's all it is. It's the plant tapping you on the shoulder and saying: something's off, and I need you — a person — to do something about it. A high bearing temperature on a shredder. A low level in a wash-water tank. A digester gas pressure drifting up.

The whole point of an alarm is that it's *actionable*. If an alarm comes in and there is nothing a human can or should do about it, it isn't really an alarm. Call it what it is: noise wearing an alarm's clothes. That distinction is going to matter a lot in a minute.

A permissive is a condition that must be true to START

A permissive (a required condition that allows a start) is a "you may proceed" gate. Before a machine will start, certain things have to be lined up. Lube oil pressure has to be established before the main drive will crank. The discharge conveyor downstream has to be running before you're allowed to start the one feeding onto it — otherwise you just pile material onto a dead belt.

A permissive doesn't stop a running machine. It only governs starting. Think of it as the checklist the logic runs through before it lets you go.

An interlock is a condition that STOPS or prevents, to protect

An interlock (a condition wired or programmed to prevent or stop motion to protect people or equipment) is the serious one. Where a permissive says "you may start," an interlock says "you will not run, or you will stop right now, because continuing is dangerous." A guard door open on a shredder. A pull-cord (the emergency stop lanyard running alongside a conveyor) pulled. A zero-speed switch saying the belt has stopped moving while the drive still thinks it's turning.

An interlock is protective by design. Its entire reason for existing is that at some moment, something more important than production is at stake.

A trip is the protective shutdown itself

When an interlock acts — when it actually drops the machine — that shutdown event is the trip (the protective stop that an interlock causes). "The crusher tripped on high vibration." The interlock is the condition and the logic. The trip is the thing happening: motor de-energized, sequence halted, often latched so it can't restart on its own until a person acknowledges it and clears the cause.

That latching is deliberate. You don't want a machine that tripped on a real fault quietly restarting the moment the condition flickers back to normal.

A safety instrumented system (SIS) is the independent safety layer

A safety instrumented system (SIS — a separate, dedicated system whose only job is to take the process to a safe state) is a layer that stands apart from the normal control PLC. On sites with combustion or gas handling — an AD plant with a boiler, a flare, or a biogas upgrading skid — the burner management or emergency shutdown function is often an SIS. It has its own sensors, its own logic solver, and its own final elements (like a slam-shut gas valve), specifically so that if the regular control system fails or is confused, the safety layer still works.

The reason it's independent is the whole idea. You don't want the same computer that might be misbehaving to also be the thing responsible for saving you from the misbehavior. Keep the safety layer separate, and one failure can't take out both.

Here's the shape of it in one breath: a permissive lets you start, an alarm asks you to act, an interlock stops you to protect, a trip is that stop happening, and an SIS is a separate guardian that doesn't trust the rest of the system to save itself.

Why alarm floods happen, and what "good" looks like

Now the practical misery: the alarm flood. That's when a single upset throws fifty alarms in two minutes and the operator can't find the three that matter. It happens because over the years, every time something went a little sideways, somebody added an alarm "so we'd know." Nobody ever went back and asked whether each one was actionable. So the screen fills with status messages, duplicates, and consequences of consequences — one trip cascades, and every downstream machine that lost its permissive alarms too.

There's real engineering guidance on this. EEMUA-191 (the Engineering Equipment and Materials Users' Association Publication 191, a widely used guide to alarm system design and management) and ISA-18.2 (the standard from the International Society of Automation covering the management of alarm systems for the process industries) both say the same core thing in different words. An alarm system is only as good as its worst hour — the upset, not the quiet shift. And they set out what a good alarm actually is:

- **Actionable.** There is a specific thing a person does in response. If not, it's not an alarm.
- **Prioritized.** High, medium, low — so that when ten come in at once, the operator's eye goes to the one that can hurt somebody or wreck a machine.
- **Timely.** It arrives with enough time left to actually do something before the situation becomes a trip. An alarm that fires half a second before the interlock does gave you nothing.

A rough gut-check the standards use: a well-run operator should be able to handle the alarms coming at them during normal running without breaking a sweat, and even in an upset the count should stay inside what a human can physically read and act on. When you're getting more alarms per minute than you can read, the system has stopped warning you and started blinding you. That's not a personnel problem. That's a design problem, and it's fixable through alarm rationalization, which is the worksheet that pairs with this primer.

The line you never cross

Here's the heart of it, and I'm going to teach it rather than preach it, because the difference is real engineering and it deserves to be understood, not just obeyed.

There is a legitimate thing you can do with a **nuisance alarm or nuisance trip**, and there is a thing that gets people killed. They can look similar from ten feet away. They are opposites.

The legitimate path: retuning a genuine nuisance. Say you've got a bearing temperature alarm set at a number that trips on a hot afternoon when the bearing is actually fine — the setpoint was picked conservatively years ago and the real safe limit, per the bearing's spec, has margin above it. That's a

nuisance: the alarm is firing inside the safe range, for no actionable reason, and it's training your operators to ignore alarms. Fixing that is good engineering. But you do it properly:

- You stay **inside the rated limit**. You move the setpoint to a value that still protects the equipment per its actual specification — you never widen it past what the equipment is rated to survive. You're removing false noise, not removing protection.
- You make **one change**, deliberately, not a handful of tweaks hoping something sticks.
- You go through **management of change (MOC — the formal process of reviewing, approving, and recording a change before it's made)**. Somebody who understands the consequence signs off.
- You **log it**. What it was, what it is now, why, who, when.

That's retuning. It makes the plant safer, because it makes the alarms mean something again.

The path that hurts people: defeating a protective interlock. Jumpering a guard-door switch so the shredder runs with the door open. Forcing a PLC input so the logic thinks a pull-cord is healthy when it's been yanked. Bypassing a gas detector because it "keeps going off." Widening a trip past the limit the equipment or the process is actually rated for. Every one of those removes protection while leaving the plant running. There is no version of that which is maintenance. It is defeating a safety function, and the reason we're all so hard about it is that the interlock only matters in the one moment you didn't expect — and that's exactly the moment you've disabled it for.

Never, ever defeat gas detection or fire detection. On an AD site especially: hydrogen sulfide (H₂S) is immediately dangerous to life or health (IDLH) around 100 ppm, and — this is the part that gets people — at high concentrations it deadens your sense of smell, so "I can't smell it anymore" can mean it's getting worse, not better. Methane is flammable between roughly 5% and 15% by volume in air (its lower and upper explosive limits, LEL/UEL). Digesters, wet wells, and gas-handling rooms can be permit-required confined spaces and are often Class I hazardous locations under the National Electrical Code (NEC 500/505). The personal 4-gas monitor you clip on before you enter (it reads oxygen, the combustible-gas LEL, carbon monoxide, and H₂S) and the fixed detection interlock are not nuisances to be silenced. They are the reason people walk back out.

The chattering interlock: a worked example

Let me make it concrete, because this is where the wrong instinct wins if you let it.

You've got an emergency-stop pull-cord along a transfer conveyor, and it keeps tripping the belt for no obvious reason. Two, three times a shift. Production's frustrated. The easy "fix" everyone's thinking is to jumper it out — force the input, prove the point, keep running.

Don't. Walk it instead. A chattering interlock is almost always telling you the truth in a garbled way. Go find the actual source:

- Is the cord over-tensioned or under-tensioned, so thermal expansion or a passing bump is enough to trip the switch? That's a mechanical adjustment at the cord.

- Is a switch box loose, corroded, or full of water so the contact is making and breaking? That's a sensor-level repair.
- Is a wiring terminal backing out and giving you an intermittent open? That's a connection to re-land.
- Is the cord physically rubbing on structure as material loads the belt? That's alignment.

Every one of those is fixable at the source, and every one of those fixes leaves the protection *intact*. The pull-cord still works when somebody actually needs it. Jumpering it out would have "solved" the chattering by removing the very thing that's supposed to stop the belt when a person is caught in it. You'd have traded a nuisance for a live plant with a dead safety function — and nobody downstream would know.

If you genuinely cannot fix it on the spot and the machine must run to avoid a worse hazard, that's a formal temporary-bypass-under-MOC situation: authorized, time-limited, documented, with compensating measures (a spotter, restricted access), and removed the moment the repair is done. That's a managed decision by people who own the risk. It is not a jumper somebody left in on night shift.

One honest word about forcing I/O

Forcing a PLC input or output — overriding what the field device says and telling the logic to see what you want it to see — is a real diagnostic tool. Sometimes you force an output to confirm a valve strokes, or check a signal path during commissioning. Used that way, deliberately, for minutes, with the plant in a known state and people clear, it's legitimate.

The danger is that a force leaves the plant in a non-normal state that doesn't look non-normal. The screen reads fine. The next person has no idea. So the rule is simple: a force is a temporary diagnostic action, it is never a fix, and it never gets left in place. If you force something, you own removing it before you walk away, and you tell the next shift.

Try this on your own plant (safely)

Pick one machine you know well. On paper, with no tools and nothing energized, sort its signals into the five buckets: which are alarms (and for each, ask honestly — what does a human *do* when this comes in?), which are permissives, which are interlocks, which cause a trip, and whether any of it belongs to a separate safety system. You'll likely find one or two "alarms" that nobody acts on — candidates for rationalization — and you'll get a much clearer picture of what's actually protecting you.

If any step would require an energized measurement — verifying a live signal at a terminal, say — treat it as the hazard it is: proper arc-flash rated personal protective equipment (PPE), the right meter, and the awareness that on any drive there's stored energy in the DC bus (the internal high-voltage capacitor bank) after power-down — wait the drive's rated discharge time, often five minutes or more,

and verify it's bled down before you trust it. For anything hands-on, do lockout/tagout (LOTO — isolating and locking off every energy source) and verify zero energy first. The paper exercise, though, you can do safely at the desk, and it's worth an hour.

A closing note

The place I see this go wrong most is where a plant's been handed between vendors and integrators over the years, and nobody carries the whole picture anymore — so alarms pile up, interlocks get "temporarily" worked around, and the meaning drains out of the system a little at a time. A second set of eyes on your alarm and interlock philosophy, from someone with no product to defend, usually earns its keep right there. Most of the work turns out to be subtraction: getting the dead alarms off the screen and the quietly worked-around interlocks back to honest, so that what's left on your operator can be trusted again.

Glossary

- **Alarm** — an annunciated abnormal condition that asks a human to take a specific action. If nothing is done in response, it shouldn't be an alarm.
- **Permissive** — a condition that must be true before a machine is allowed to *start*. Governs starting only, not running.
- **Interlock** — a condition, wired or programmed, that prevents or stops motion to protect people or equipment.
- **Trip** — the protective shutdown itself, the event of an interlock dropping the machine.
- **Safety instrumented system (SIS)** — a separate, dedicated system with its own sensors, logic, and final elements whose only job is to bring the process to a safe state, independent of the normal control PLC.
- **PLC (programmable logic controller)** — the industrial computer that runs the plant's control sequence.
- **Alarm flood** — many alarms arriving faster than an operator can read and act on them, usually triggered by one upset cascading.
- **EEMUA-191** — Engineering Equipment and Materials Users' Association Publication 191, a widely used guide to alarm system design and management.
- **ISA-18.2** — International Society of Automation standard for the management of alarm systems in the process industries.
- **Alarm rationalization** — the structured review of every alarm to confirm it's actionable, correctly prioritized, and timely.
- **Nuisance alarm / nuisance trip** — one that fires inside the safe range for no actionable reason, training people to ignore the system.

- **Management of change (MOC)** — the formal process of reviewing, approving, and recording a change before it's made.
- **Forcing I/O** — overriding a field signal in the PLC; a legitimate short-term diagnostic action, never a fix, never left in place.
- **Pull-cord** — the emergency-stop lanyard running alongside a conveyor.
- **4-gas monitor** — the personal instrument worn before entering a confined or gas-handling space, typically reading oxygen, combustible-gas LEL, carbon monoxide, and H₂S.
- **H₂S (hydrogen sulfide)** — toxic gas, roughly 100 ppm IDLH; deadens the sense of smell at high concentration.
- **IDLH** — immediately dangerous to life or health.
- **LEL / UEL** — lower and upper explosive limits; methane is flammable at roughly 5%–15% by volume in air.
- **NEC 500/505** — National Electrical Code articles covering hazardous (classified) locations, e.g. Class I where flammable gas may be present.
- **PPE** — personal protective equipment, including arc-flash rated gear for energized work.
- **DC bus** — the internal high-voltage capacitor bank in a drive that holds stored energy for minutes after power-down.
- **LOTO (lockout/tagout)** — isolating and locking off every energy source, then verifying zero energy, before hands-on work.